

Vincent S. Pan

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Website: <https://vsbpan.github.io/> | GitHub: <https://github.com/vsbpan>

EDUCATION

- 2021 – 2026 **Ph.D., Integrative Biology | Ecology, Evolution, & Behavior**
Michigan State University, East Lansing, MI
Thesis co-advisors: William C. Wetzel, Kadeem J. Gilbert
- 2017 – 2020 **B.S., Animal Biology** *summa cum laude*
University of California- Davis, Davis, CA
Thesis advisor: Rick Karban
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EXPERIENCE

- 2026 – Present **Postdoctoral Fellow** University of South Carolina, Columbia, SC
Supervisor: Tad A. Dallas
- 2025 – 2025 **Visiting Scholar** University of South Carolina, Columbia, SC
Host PI: Eric F. LoPresti
- 2023 – 2024 **Visiting Scholar** Colorado State University, Fort Collins, CO
Host PI: Ian S. Pearse, Paul J. Ode
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PUBLICATIONS

*Undergraduate mentee; †Co-first authors

17. Gaughan, M.^{*†}, **Pan, V. S.[†]**, Martin-Eberhardt, S., Wetzel, W. C., Gilbert, K. J. 2026. Herbivory Weakens the Stability of a Floral Polymorphism. *The American Naturalist* In Press.
16. Karban, R., Pepi, A. A., Grof-Tisza, P., **Pan, V. S.**, Loeb, G., Huntzinger, M., Holyoak, M. 2026. Precipitation Drives the Abundance and Distribution of *Arctia virginalis*: A 40-Year Study. *BioScience* 76:375–384.
15. **Pan, V. S.**, Gilbert, K. J., Wetzel, W. C. 2025. Mean Plant Toxicity Modulates the Effects of Plant Defense Variability. *Ecology* 106:e70012.
14. **Pan, V. S.**, Ghosh, E., Ode, P. J., Wetzel, W. C., Gilbert, K. J., Pearse, I. S. 2025. Large Differences in Herbivore Performance Emerge from Simple Herbivore Behaviors and Fine-Scale Spatial Heterogeneity in Phytochemistry. *Ecology Letters* 28:e70044.
13. **Pan, V. S.**, Wetzel, W. C. 2024. Neutrality in Plant–Herbivore Interactions. *Proceedings of the Royal Society B: Biological Sciences* 291:20232687.
12. Robinson, M. L., . . . , **Pan, V. S.**, . . . , Wetzel, W. C. 2023. Plant Size, Latitude, and Phylogeny Explain Variability in Global Herbivory. *Science* 382:679-683.

11. Pepi A, **Pan, V. S.**, Grof-Tisza, P., Holyoak, M., Balldman, A., Laws-McNeil, A., Mase, V., Moseley, C., Karban, R. 2023. Spatial Habitat Heterogeneity Influences Host-Pathogen Dynamics in a Patchy Population of Ranchman's Tiger Moth. *Ecology* 104:e4144.
10. LoPresti, E. F., **Pan, V. S.**, Girvin, C., Barber, G., Jaeger, S., Orrock, J. 2023. Sandy Seeds: Armor or Invisibility Cloak? Mucilage-Bound Sand Physically Protects Seeds From Rodents and Invertebrates. *Journal of Ecology* 111:540-551.
9. **Pan, V. S.**, Pepi, A., LoPresti, E. F., Karban, R. 2023. The Consequence of Leaf Lifespan to Virus Infection of Herbivorous Insects. *Oecologia* 201:449-459.
8. Pepi, A., **Pan, V. S.**, Rutkowski, D., Mase, V., Karban, R. 2022. Influence of Delayed Density and Ultraviolet Radiation on Caterpillar Baculovirus Infection and Mortality. *Journal of Animal Ecology* 91:2192–2202.
7. Orrock J. L., Guiden, P. W., **Pan, V. S.**, Karban, R. 2022. Plant Induced Defenses That Promote Cannibalism Reduce Herbivory as Effectively as Highly Pathogenic Herbivore Pathogens. *Oecologia* 199:397–405.
6. **Pan, V. S.**, Girvin, C., LoPresti, E. F. 2022. Anchorage by Seed Mucilage Prevents Seed Dislodgement in High Surface Flow: A Mechanistic Investigation. *Annals of Botany* 129:817–830.
5. **Pan, V. S.**, Marshall, M., Karban, R., Goidell, J., Weber, M. G., LoPresti, E. F. 2021. Mucilage-Binding to Ground Protects Seeds of Many Plants From Harvester Ants: A Functional Investigation. *Functional Ecology* 35:2448–2460.
4. **Pan, V. S.**, Pepi, A., Goidell, J., Karban, R. 2021. Retention of Green Leaves Not Brown Leaves Increases Spring Cynipid Diversity on Large Valley Oaks. *Arthropod-Plant Interactions* 15:353–362.
3. Grof-Tisza, P., Karban, R., **Pan, V. S.**, Blande, J. D. 2020. Assessing Plant-to-Plant Communication and Induced Resistance in Sagebrush Using the Sagebrush Specialist *Trirhabda pilosa*. *Arthropod-Plant Interactions* 14:327–332.
2. Nguyen, H. D. D., **Pan, V. S.**, Pham, C., Valdez, R., Doan, K., Nansen, C. 2020. Night-Based Hyperspectral Imaging to Study Association of Horticultural Crop Leaf Reflectance and Nutrient Status. *Computers and Electronics in Agriculture* 173:105458.
1. LoPresti, E. F., **Pan, V. S.**, Goidell, J., Weber, M. G., Karban, R. 2019. Mucilage-Bound Sand Reduces Seed Predation by Ants but Not by Reducing Apparency: A Field Test of 53 Plant Species. *Ecology* 100:e02809.

Preprints or manuscripts in review

- P5. **Pan, V. S.**, Sedio, B. E., Pearse, I. S. A universal metabolite occupancy spectrum reveals the structure of plant chemodiversity.
- R4. **Pan, V. S.**, Adam, J., Anstett, D., Carvajal Acosta, N., Cornelissen, T., Galmán, A., Haslup, P., Karp, J., López-Goldar, X., Martin-Eberhardt, S., Ritter, K., Santos Lopes, H. D., Wonderlin, N., Wetzal, W. C. 2026. Nonlinear Impacts of Herbivory on Plants Explain the Herbivory Paradox. *bioRxiv*. <https://doi.org/10.64898/2026.02.24.707788>

- R3. **Pan, V. S.**, Rothstein, P. E. *, Gilbert, K. J. 2026. The Frequency of Population and Masting Fluctuations Is Increasing Globally. *bioRxiv*. <https://doi.org/10.64898/2026.02.04.701112>
- R2. Martin-Eberhardt, S., **Pan, V. S.**, Gilbert, K. J., Weber, M. 2026. Convergent evolution of red pigmentation in extrafloral nectaries: global patterns and mechanisms. *The American Naturalist* Major revisions.
- R1. Ghosh, E., **Pan, V. S.**, Pearse, I. S., Ode, P. J. Herbivore Gains Immunotherapeutic Effects From Low Variation in Plant Defense Toxins. *Journal of Chemical Ecology* Minor revisions.

Manuscripts in preparation

- P2. Nagarkar, S. M., Curley, S. J., **Pan, V. S.**, LoPresti, E. F., Szczepanski, C. R., An Engineered Approach to Characterize Mucilage Adhesion in Diverse Species and Environments.
- P1. LoPresti, E. F., **Pan, V. S.**, Baskett, C., Dallas, T., Mason, C., Nerhkorn, C., Weber, M. G. A Geometric Model Explains Repeated Evolution of the World's Most Common Floral Form.

PRESENTATIONS

Invited Talks

- Pan, V. S.**, Gaughan, M., Martin-Eberhardt, S., Wetzel, W. C., Gilbert, K. J. 2024. Early Season Herbivory Stabilizes a Floral Polymorphism: The Science and the People Behind the Science. *In: The future is here: Undergraduate-driven research in a changing world. Botany*. Grand Rapids, MI, USA.
- Pan, V.S.**, Pearse, I.S., Gilbert, K.J., Adam, J., Anstett, D., Carvajal, N., Cornelissen, T., Galmán, A., Ghosh, E., Haslup, P., Karp, J., López-Goldar, X., Martin-Eberhardt, S., Ode, P., Ritter, K., Santos Lopes, H. D., Wonderlin, N., Wetzel W.C. 2025. Emergent Properties of Stochastic Plant-Herbivore Interactions. University of South Carolina EEB seminar, Columbia, SC, USA.

Contributed Talks

- Pan, V. S.**, Ghosh, E., Ode, P. J., Wetzel, W. C., Gilbert, K. J., Pearse, I. S. 2024. Spatially Heterogeneous Phytochemical Landscapes Increase Herbivore Performance by Enhancing Herbivore Foraging. *Ecological Society of America Annual Meeting*. Long Beach, CA, USA.
- Pan, V. S.**, Wetzel, W. C. 2022. A Neutral Model of Herbivory: fundamental constraints on patterns and variation. *Ecological Society of America Annual Meeting*. Montreal, QC, Canada
- Pan, V. S.**, Pepi, A., LoPresti, E. F., Karban, R. 2021. Leaf Retention Increases Infection Risk of an Arctia Caterpillar. *Arctiinae Research Community Meeting*. Virtual.

Posters

- Pan, V. S.,** Wetzel, W. C. 2023. A Neutral Model of Herbivory: Fundamental constraints on patterns and variation. *Gordon Research Conference- Plant Herbivore Interaction*. Ventura, CA, USA.
- Pan, V. S.,** Beebe, O., Wetzel, W. C. 2023. Causes and Consequences of Within Leaf Spatial Aggregation of Herbivory. *Gordon Research Seminar- Plant Herbivore Interaction*. Ventura, CA, USA.
- Pan, V. S.,** Adam, J., Anstett, D., Carvajal Acosta, N., Cornelissen, T., Galmán, A., Haslup, P., Karp, J., López-Goldar, X., Martin-Eberhardt, S., Ritter, K., Santos Lopes, H. D., Wonderlin, N., Wetzel, W. C. 2025. Does herbivory matter? A meta-analysis of nonlinearity in tolerance to herbivore damage. *Gordon Research Conference- Plant Herbivore Interaction*. Pomona, CA, USA.

SOFTWARE

- Pan, V. S.** 2023. *herbivar*: Quantitative Tools for Variance Explicit Ecology and Herbivory. R package version 0.2.0. <https://github.com/vsbpan/herbivar>

AWARDS

External

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| 2023 | National Science Foundation INTERN Program (DEB 2203582) (\$37,982) |
| 2021 | National Science Foundation Graduate Research Fellowship (\$138,000) |

Internal

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| 2024 | Plant Resilience Institute Travel Award (\$500) |
| 2023 | Kellogg Biological Station Mentoring Award (\$1,500) |
| 2023 | John R. Shaver Research Fellowship (\$2,016) |
| 2022 – 2024 | Ecology, Evolution, and Behavior Professional Horizons Grant (\$1,750) |
| 2022 | Kellogg Biological Station Fellowships for Graduate Students (\$3,750) |
| 2021 | Michigan State University- University Distinguished Fellowship (\$78,000) |
| 2021 | Michigan State University- Early Start Fellowship (\$6,000) |
| 2021 | Michigan State University- Plant Science Fellowship (\$2,500) |

MENTORSHIP

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| 2025 – 2025 | Peter Rothstein (<i>Research Experience for Undergraduates</i>)
(Poster) <i>Increasing Frequency of Population Fluctuations</i> , Summer Research Symposium, Kellogg Biological Station, Michigan State University. |
| 2023 – 2026 | Mia Gaughan (<i>Research Experience for Undergraduates</i>)
(Poster) <i>Rarity Increases Herbivory of a Polymorphic Plant</i> , Summer Research Symposium, Kellogg Biological Station, Michigan State University. |

(Invited talk) Early Season Herbivory Stabilizes a Floral Polymorphism: The Science and the People Behind the Science *In: The future is here: Undergraduate-driven research in a changing world. Botany 2024.*

2022 – 2023 **Olivia Beebe (*Undergraduate Research Apprenticeship*)**

(Poster) *Sticky Situations: Assessing seed mucilage of Aizoaceae*, Summer Research Symposium, Kellogg Biological Station, Michigan State University.

(Poster) *Sticky Situations: Assessing seed mucilage of Aizoaceae*, University Undergraduate Research and Arts Form, Michigan State University.

TEACHING

Courses

2024 – 2025 **Introduction to data analysis in R**
2 semesters Co-instructor, designed new curriculum for 8-week course
 Kellogg Biological Station summer student course (~20 undergraduate students)

2022, 2024 **The practice of science reading group**
2 semesters Organizer
 Michigan State University (7 – 10 undergraduate and graduate students, staff)

2023 **Introduction to the R software for ecologists**
1 semester Co-instructor
 Kellogg Biological Station summer student course (~20 undergraduate students)

Guest lectures

2023 Getting Started in undergraduate research at Nanjing University

REVIEW

Oecologia (2022), *Proc. B* (2022), *BMC Res. Notes* (2023), *Am. J. Bot.* (2023), *Mol. Ecol.* (2024), *J. Pest Sci.* (2025), *Ann. Bot.* (2025), *Ecosphere* (2025), *Phil. Trans. B* (2025, 2026) *Ecology* (2026), *APIS* (2026)

SERVICE

2025 Envision EEB Graduate School Funding Workshop

2025 Peer Mentor Program Co-chair

2023 Plant Science Fellowship Recruitment Committee

2022 – 2023 Ecology, Evolution, & Behavior Student Colloquium Co-chair
